

# JEROME ANTONY ROBIN

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## SUMMARY

Electronics and Communication Engineering student specializing in digital ASIC/FPGA acceleration, bare-metal runtime architectures, and low-level Linux systems programming. Proven record designing high-throughput systolic compute engines, building automated open-source RTL-to-GDSII environments, and deploying optimized edge AI pipelines across Microchip PolarFire, Xilinx Zynq, and NVIDIA Jetson platforms.

## TECHNICAL SKILLS

- **Languages:** C, C++ (C++17/20), Python, Verilog, VHDL, ARM & MIPS Assembly, Bash Scripting
- **Silicon & FPGA:** RTL Design, Microchip Libero SoC, Synplify Pro, Xilinx Vivado/HLS, Open-Source IC Design (OpenROAD, Yosys, OpenSTA, Magic VLSI, GHDL), RISC-V Architecture (RV32I/M, NEORV32)
- **Bus Protocols & Interfaces:** AXI4, AXI4-Lite, AXI4-Stream, AXI-DMA, Wishbone, UART, SPI, I2C
- **Embedded & Systems:** Linux Userspace Drivers & VFS (V4L2), Physical Memory Mapping, Bare-Metal Runtime Development, RP2040 Pico SDK, Dynamic Memory Arenas, pybind11, CMake, Git / GitHub
- **Binary Engineering & Runtimes:** Dynamic ELF Patching, Android Bionic-to-Glibc Translation Shims
- **Hardware Edge AI:** INT8 / FP16 Quantization, Hardware GEMM Units, BNN Acceleration, TensorRT, Vitis AI, TFLite Micro, ONNX, OpenCV

## EDUCATION

### Easwari Engineering College

**Bachelor of Engineering in Electronics and Communication Engineering**

2024 – 2028 | CGPA: 7.6

## AWARDS & HONORS

- **1st Prize** — TechPulse 2026, Easwari Engineering College (Project Presentation)
- **2nd Prize** — Nexus Chipcraft 2026, RMKCET (EDA / RTL Design Track)
- **3rd Prize** — Vidyutrenz 2026, CIT (Embedded Systems & Acceleration)

## WORK EXPERIENCE

### e-con Systems

#### Edge AI & Embedded Vision Intern

*April 2026 – November 2026 (7 mos) | Chennai, India*

- Engineered a container-free native Linux runtime port of Google's mobile vision engine for x86\_64 and ARM64 platforms via a custom Bionic-to-glibc translation shim.
- Developed high-throughput, zero-copy C++ and Python SDKs interfacing multi-camera V4L2 video streams directly with embedded neural network backends.
- Trained, pruned, and fine-tuned edge vision models using PyTorch; executed post-training quantization (INT8/FP16) and deployed optimized pipelines across TensorRT for NVIDIA Jetson and TFLite for edge devices.

### Directorate of Research, SRM Group of Institutions

#### Research Intern (FPGA & Systems)

*January 2026 – Present (10 mos) | Chennai, India*

- Developed custom hardware accelerators and compute pipelines using Verilog, VHDL, and Vivado HLS across Xilinx Zynq and Microchip PolarFire FPGA architectures.
- Ran cycle-accurate RTL verification, testbench simulation, and corner-case functional debug using Vivado, GHDL, and ModelSim.
- Integrated and evaluated on-chip bus interconnects and protocols, handling AXI4-Lite, AXI-Stream, AXI-DMA, and Wishbone interfaces.
- Built bare-metal C/C++ runtimes and TFLite Micro inference engines on ARM Cortex-M microcontrollers, implementing static memory arena management.
- Performed physical board-level validation, timing debug, and hardware-software co-design using logic analyzers, /dev/mem drivers, and Linux interfaces.

### The Silis Foundation

#### Lead Developer & Architect

*December 2025 – Present (11 mos)*

- Architected Silis to democratize custom silicon design, rejecting fragmented proprietary monoliths in favor of an open, deterministic RTL-to-GDSII workbench that makes physical IC design accessible to beginners.
- Unified open-source synthesis, place-and-route, timing analysis, and DRC/GDS engines into an automated execution plane using CMake, modern C++, PyQt6, and Python.

## KEY PROJECTS

### PulseGEMM: PolarFire Systolic Engine for CNNs | [GitHub](#)

*Verilog, Libero SoC, Synplify Pro, C*

- Designed an INT8 16×16 processing-element (PE) systolic array core on Microchip PolarFire Discovery SoC Kit, driving 256 hardware DSP blocks (87.7% device budget) to deliver 256 MACs/cycle at 100 MHz.
- Built a 64-bit AXI4 master DMA engine with ping-pong SRAM buffering, overlapping compute with memory transfers to execute zero-copy 2D strided address extraction from activation maps.
- Developed low-level /dev/mem C hardware drivers (gemmrv\_hw), clocking a 51.58x end-to-end speedup over CPU execution on ResNet-50 Conv layers.

### GEMMrv32: RISC-V SoC with Custom Matrix Engine | [GitHub](#)

*VHDL, Verilog, NEORV32, GHDL, C*

- Extended a 32-bit pipelined RISC-V processor (rv32im) with an 8×8 systolic-array compute engine (64 parallel PEs) integrated into the Custom Functions Subsystem (CFS).
- Demonstrated cycle-accurate execution in GHDL, achieving a 26.2x speedup over an optimized software baseline (-O3 with loop unrolling) for 16×16 matrix workloads.

### RP2040 Distributed Compute Cluster & Execution Platform | [GitHub](#)

*C, ARM Cortex-M0+ Assembly, Pico SDK, UART*

- Engineered an 8-core distributed computing architecture across 4 networked RP2040 nodes with 1 MB aggregate non-shared SRAM for parallel pipeline and embedded AI execution.
- Implemented a bare-metal runtime workload distributor and custom packetized, interrupt-driven UART protocol for low-latency inter-node communication and synchronization.
- Integrated dynamic binary execution routines with standalone bitmapped VGA output and USB human interface device (HID) peripherals to operate as a self-contained computer

### Binarized YOLO on PYNQ Fabric | [GitHub](#)

*Vivado HLS, C++, Python, PYNQ-Z2*

- Modernized overlay driver pipelines for PYNQ v2.7+ (Ubuntu 20.04), migrating legacy Xlnk drivers to contiguous pynq.allocate DMA memory buffers.
- Achieved a ~150x speed multiplier over edge ARM CPU execution for real-time 20-class object detection.

### tinymml-rt-pico: Dynamic Edge AI Runtime | [GitHub](#)

*C++, ARM Cortex-M0+ Assembly, Pico SDK*

- Built a bare-metal execution runtime decoupling neural networks from MCU firmware by hot-swapping .tflite models and pipeline configs from an SPI SD card without reflashing.
- Engineered zero-buffer memory arena management to dynamically swap tensor execution graphs within the strict 264 KB SRAM budget of the RP2040.